



Zomato Restaurant Review – Insights Report



Zomato Restaurant Review Analysis & Prediction



Project Overview

This project performs an **end-to-end data analysis and machine learning workflow** on the Zomato Restaurant Review dataset.

The objective is to clean and analyze restaurant and review data, validate assumptions using statistical tests, engineer meaningful features, and build machine learning models to predict restaurant cost categories and ratings.



Dataset Description

The dataset consists of **restaurant-level** and **review-level** information.

Restaurant Information

- `Name` – Restaurant name
- `Links` – Zomato restaurant URL
- `Cost` – Average cost for two people
- `Collections` – Zomato collections the restaurant belongs to
- `Cuisines` – Types of cuisines offered
- `Timings` – Restaurant operating hours

Review Information

- `Restaurant` – Restaurant name (review reference)
 - `Reviewer` – Reviewer identifier
 - `Review` – Customer review text
 - `Rating` – Customer rating
 - `Metadata` – Review metadata
 - `Time` – Review timestamp
 - `Pictures` – Review images availability
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Project Workflow

1 Data Loading

- Loaded dataset using **Pandas**
 - Verified shape, column names, and data types
 - Performed initial sanity checks
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2 Dataset Information

- Combination of **numerical, categorical, text, and datetime** features
 - Enables both **structured analysis** and **NLP-based modeling**
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3 Understanding Variables

- `Cost` and `Rating` act as target variables
 - `Cuisines` and `Collections` represent restaurant segmentation
 - `Review` captures customer sentiment
 - `Time` supports temporal analysis
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4 Unique Value Analysis

- Checked unique values per column to understand cardinality
 - Identified high-cardinality fields (`Restaurant`, `Reviewer`)
 - Confirmed categorical suitability for modeling (`Rating`, `Cost_Category`)
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5 Data Preprocessing

- Converted numerical columns (`Cost`, `Rating`) to numeric types
 - Handled missing and duplicate values
 - Normalized text data (lowercasing, trimming whitespace)
 - Parsed datetime fields from `Time`
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6 Data Wrangling

- Split and exploded multi-valued columns (`Cuisines`, `Collections`)
 - Standardized categorical values
 - Aggregated restaurant-level and review-level data where required
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7 Data Visualization

Used **Matplotlib** and **Seaborn** for EDA:

- Cost distribution histograms
 - Box plots of cost by cuisine and collection
 - Bar plots of average cost by cuisine
 - Correlation heatmap using engineered numerical features
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8 Hypothesis Testing

Conducted statistical tests to validate assumptions:

- **One-Way ANOVA**
 - Tested whether restaurant cost differs across cuisine types
 - Result: Statistically significant differences observed ($p < 0.05$)
 - **T-Test / ANOVA**
 - Compared costs across collection categories
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9 Feature Engineering

Created new features to improve analysis and modeling:

- Cuisine_Count
 - Collection_Count
 - Name_Length
 - Timings_Length
 - Cost_Category (Budget / Mid-range / Premium)
 - TF-IDF features extracted from review text
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10 Machine Learning Model Implementation

Models Used:

- **Linear Regression** – Cost prediction
- **Logistic Regression** – Cost category / rating classification

Workflow:

1. Feature selection and encoding
2. Train-test split
3. Feature scaling
4. Model training using `.fit()`

5. Prediction using `.predict()`
 6. Performance evaluation
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Model Prediction

- Predicted cost and rating categories for unseen restaurants
 - Demonstrated model generalization and usability for business insights
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Model Evaluation Metrics

- **Regression:** R^2 Score, MAE, RMSE
 - **Classification:** Accuracy, Precision, Recall, F1-Score
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Key Insights

- Restaurants offering multiple cuisines generally have higher costs
 - Premium collections show significantly higher median pricing
 - Review sentiment contributes to rating prediction
 - Pricing varies significantly across cuisine categories
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Business Impact

- Enables data-driven restaurant segmentation
 - Supports pricing and recommendation strategies
 - Helps understand customer sentiment and preferences
 - Useful for restaurant owners and food-tech platforms
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Tools & Technologies

- Python
 - Pandas, NumPy
 - Matplotlib, Seaborn
 - Scikit-learn
 - NLP (TF-IDF)
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Future Enhancements

- Advanced NLP models (Word2Vec, BERT)
 - Tree-based models (Random Forest, XGBoost)
 - Interactive dashboards (Power BI / Tableau)
 - Sentiment-based recommendation system
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